What if Mass Eats the Vacuum?

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Abstract

This paper introduces a classical field theory of gravity built on the Maxwell equations and a hypothesis: that mass absorbs radiation, inducing a net inward momentum flow. We assume a Lorentz-invariant stochastic zero-point radiation field present throughout space and hypothesize that mass acts as a sink for this field. We construct a consistent model of a massive particle, demonstrate energy and momentum conservation, and show that the three classic tests of general relativity: Mercury's precession, gravitational lensing, and gravitational redshift are reproduced to leading order. This approach provides a novel, Lorentz-invariant, classical alternative to general relativity, and lays a foundation for further exploration.

1 Introduction

Modern physics rests on two towering achievements: the Standard Model, underpinned by the Dirac equation, and General Relativity, based on Einstein's field equations. One governs the small, the other the large. Each is remarkably successful in its domain.

These two frameworks, however, are incompatible. When applied simultaneously, the equations break down and yield infinities. Proposals like string theory offer elegant mathematics but few concrete predictions. Despite nearly a century of effort, no unifying framework has emerged. A fresh approach is warranted.

The scientific method permits us to begin with a hypothesis. Newton postulated an inverse square force law without explaining why gravity falls off with distance. Maxwell wrote down his equations before anyone understood the photon. A hypothesis does not need to come with a microscopic derivation. A theory only needs to make testable predictions. That is the standard we hold ourselves to here.

This paper proposes such an approach. Rather than quantizing Einstein's field equations or geometrizing the Dirac equation, we suggest something simpler and stranger: what if mass eats the vacuum?

1.1 Roadmap

This paper is structured as a straightforward application of the scientific method: we begin with a hypothesis, outline the assumptions that support it, explore its mathematical consequences, compare its predictions to experiment, and finally examine whether it motivates a deeper conceptual shift.

In **Section 2**, we introduce the hypothesis: that mass acts as a sink for radiation and, in particular, as a sink for a pervasive stochastic zero-point radiation field (Z.P.R.). We first present the idea intuitively, anchoring the theory in a physical picture, and then spell it out mathematically.

In **Sections 3 and 4**, we lay out the two assumptions underlying the theory. By "assumptions," we mean elements already present in the scientific literature, perhaps unconventional but not novel to this work. **Section 3** introduces the stochastic zero-point radiation (Z.P.R.) field as the physical manifestation of the vacuum. **Section 4** adopts the Maxwell equations as the correct mathematical framework for describing gravity.

Sections 5 and 6 contain the technical core of the paper. We introduce a toy model of a massive particle and discuss characteristics of the toy model. We examine whether the inflow of radiation can stabilize mass against gravitational self-energy and show that the energy needed for stability of our toy model is less than that available in the Z.P.R. We also demonstrate conservation of energy and momentum and several other key mathematical results.

In **Section 7**, we test the theory against observation. We demonstrate that it reproduces the three classic predictions of general relativity to leading order:

perihelion precession, gravitational lensing, and redshift. We also identify a testable difference between general relativity and the theory presented here.

Section 8 explores a surprising result: when the field equations are rewritten in spinor form, they yield a structure identical to the Dirac equation, without quantization. And finally, **Section 9** concludes with a discussion of open questions, potential avenues for further investigation, and the broader implications of the framework proposed here.

2 The Hypothesis

In this section, we introduce the hypothesis of this paper. Before proceeding, we clarify the distinction between assumptions and hypothesis as used here:

Assumptions are drawn from existing theoretical frameworks, even if unconventional. While we do not attempt to derive or fully justify these assumptions from first principles, we do examine their consequences — particularly in Section 7, where contact with experiment is made. The **hypothesis** refers to the novel claim proposed in this work.

Here we first present the hypothesis intuitively, anchoring the theory in a physical picture, and then spell it out mathematically.

2.1 A Physical Picture: Mass as a Radiation Sink

Before introducing equations, we present a simple physical picture to guide intuition. Consider a neutral, perfectly conducting iron sphere, heated until it glows red. When placed in the vacuum of space, it emits infrared radiation and cools down, approaching absolute zero.

But at zero Kelvin, both classical and quantum theories suggest the presence of a background electromagnetic field called the stochastic zero-point radiation (Z.P.R.) field. A fuller treatment of the Z.P.R. will be offered in **Section 3**, but for now, it is enough to imagine a low-level radiation noise present even in the coldest vacuum. The iron sphere, after it has cooled down and come into equilibrium with the vacuum, then absorbs radiation from this ambient field.

The hypothesis proposed here is that mass absorbs energy from the zero-point field even at absolute zero. This interaction produces a net inward flow of momentum toward the mass.

The remainder of the paper builds from this intuition and develops the consequences of this idea.

2.2 Hypothesis: The Math

The hypothesis of this paper is that mass creates an inward-pointing spectral momentum density:

$$\boxed{\mathbf{P}(r) = -\hat{\mathbf{r}}\frac{Gm\hbar}{r^2}} \tag{1}$$

A photon traveling through this field experiences a cumulative momentum shift:

$$\Delta\mathbf{p} = \int_{\text{path}} \mathbf{P}(r)\, dl \tag{2}$$

The intuition underlying this hypothesis has respectable precedent. Feynman notes that light "falls" in a gravitational field [?], which is precisely the picture developed here: mass acts as a sink for radiation. We do not specify the microscopic mechanism by which mass couples to the ZPR field. The implications of this hypothesis are tested in **Section 7**.

3 Zero Point Radiation (ZPR)

We now introduce our first assumption: the **stochastic zero-point radiation field**. "Stochastic" means random, in a way that will be clearly defined below, "zero-point" refers to zero-temperature and "radiation" in this context simply means light.

Historically, the Michelson–Morley experiment aimed to detect the luminiferous aether, the medium through which light was thought to propagate. The null result helped solidify the notion of a truly empty vacuum. In contrast, the stochastic zero-point radiation field treats the vacuum as filled with fluctuating electromagnetic fields, even at absolute zero.

This theory was first developed by T.H. Boyer last century but has more interest as of late under the term stochastic electrodynamics (S.E.D.). Stochastic zero-point radiation has been shown to reproduce:

- The Planck blackbody spectrum (Boyer, 1969) [?]
- Harmonic oscillator ground-state energy $\frac{1}{2}\hbar\omega$ [?]
- Casimir and van der Waals forces [?]
- Certain features of the hydrogen atom's ground state [?]
- The Unruh effect [?]

Taken together, these results support the view, held by Boyer and others, that the vacuum is an active medium, filled with fluctuating modes that persist even at $T = 0$. This is not the mainstream view of the vacuum. We adopt it nonetheless, on the grounds that it has a substantial body of supporting results and, as we will show, leads to a consistent and predictive theory of gravity.

We therefore assume that the vacuum is filled with a classical, Lorentz-invariant random electromagnetic radiation field even at absolute zero, carrying a spectral energy density:

$$\rho(\omega) = \frac{1}{2}\hbar\omega \tag{3}$$

3.1 Mathematical Definitions of the Z.P.R.

The Z.P.R. field is stochastic and Gaussian. The electric and magnetic fields are defined as:

$$
\begin{aligned}
\mathbf{E}(\mathbf{x},t) &= \sum_{\lambda=1}^{2}\int d^3k\,\hat{\epsilon}(\mathbf{k},\lambda)\sqrt{\frac{\hbar\omega}{8\pi^2}}W_\lambda(\mathbf{k})e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} + \text{c.c.} \\
\mathbf{B}(\mathbf{x},t) &= \sum_{\lambda=1}^{2}\int d^3k\,\frac{\mathbf{k}\times\hat{\epsilon}(\mathbf{k},\lambda)}{|\mathbf{k}|}\sqrt{\frac{\hbar\omega}{8\pi^2}}W_\lambda(\mathbf{k})e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} + \text{c.c.}
\end{aligned}
\tag{4}
$$

where $\hat{\epsilon}(\mathbf{k},\lambda)$ are polarization vectors satisfying:

$$
\hat{\epsilon}\cdot\mathbf{k}=0,\quad \hat{\epsilon}_1\cdot\hat{\epsilon}_2=0,\quad |\hat{\epsilon}_\lambda|=1 \tag{5}
$$

and $W_\lambda(\mathbf{k})$ are complex Gaussian random variables with correlation properties:

$$
\begin{aligned}
\langle W_\lambda(\mathbf{k})\rangle &= 0 \\
\langle W_\lambda(\mathbf{k})W^*_{\lambda'}(\mathbf{k}')\rangle &= \delta_{\lambda\lambda'}\delta^3(\mathbf{k}-\mathbf{k}') \\
\langle W_\lambda(\mathbf{k})W_{\lambda'}(\mathbf{k}')\rangle &= 0
\end{aligned}
\tag{6}
$$

where $\langle\cdot\rangle$ denotes the stochastic average over the ensemble of random field realizations. This background field is homogeneous, isotropic, and statistically stationary, with the full ensemble remaining invariant under Lorentz transformations.

3.2 Derivation of Energy Spectrum

We now verify that this field definition gives the correct energy spectrum. We need to evaluate $\langle\mathbf{E}^2\rangle$ and $\langle\mathbf{B}^2\rangle$. Let us focus on $\langle\mathbf{E}^2\rangle$ as the calculation for the magnetic field is identical.

Write $\mathbf{E}=\mathbf{E}_+ + \mathbf{E}_+^*$ where $\mathbf{E}_+$ is the positive-frequency part (containing W_λ but not W_λ^*). Then:

$$
\mathbf{E}^2 = (\mathbf{E}_+ + \mathbf{E}_+^*)^2 = \mathbf{E}_+^2 + 2\mathbf{E}_+\cdot\mathbf{E}_+^* + (\mathbf{E}_+^*)^2 \tag{7}
$$

Taking the stochastic average:

$$
\langle\mathbf{E}^2\rangle = 2\langle\mathbf{E}_+\cdot\mathbf{E}_+^*\rangle \tag{8}
$$

The $\mathbf{E}_+^2$ and $(\mathbf{E}_+^*)^2$ terms vanish because $\langle WW\rangle=0$, by equation (6). Evaluating the remaining term:

$$
\langle\mathbf{E}_+\cdot\mathbf{E}_+^*\rangle = \sum_{\lambda,\lambda'}\int d^3k\,d^3k'\,\hat{\epsilon}_\lambda\cdot\hat{\epsilon}^*_{\lambda'}\frac{\hbar\sqrt{\omega\omega'}}{8\pi^2}\langle W_\lambda(\mathbf{k})W^*_{\lambda'}(\mathbf{k}')\rangle e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}e^{-i(\mathbf{k}'\cdot\mathbf{x}-\omega' t)} \tag{9}
$$

Using the correlation function $\langle W_\lambda(\mathbf{k})W^*_{\lambda'}(\mathbf{k}')\rangle = \delta_{\lambda\lambda'}\delta^3(\mathbf{k}-\mathbf{k}')$:

$$\langle \mathbf{E}_+ \cdot \mathbf{E}_+^* \rangle = \sum_\lambda \int d^3k \, |\hat{\epsilon}_\lambda|^2 \frac{\hbar\omega}{8\pi^2} \tag{10}$$

Since $\sum_{\lambda=1}^{2} |\hat{\epsilon}_\lambda|^2 = 2$ (two orthogonal unit polarization vectors):

$$\langle \mathbf{E}_+ \cdot \mathbf{E}_+^* \rangle = \int d^3k \, \frac{\hbar\omega}{4\pi^2} \tag{11}$$

Therefore:

$$\langle \mathbf{E}^2 \rangle = 2 \int d^3k \, \frac{\hbar\omega}{4\pi^2} \tag{12}$$

By identical calculation:

$$\langle \mathbf{B}^2 \rangle = 2 \int d^3k \, \frac{\hbar\omega}{4\pi^2} \tag{13}$$

The total energy density is:

$$\langle u \rangle = \frac{1}{8\pi}(\langle \mathbf{E}^2 \rangle + \langle \mathbf{B}^2 \rangle) = \frac{1}{8\pi} \cdot 4 \int d^3k \, \frac{\hbar\omega}{4\pi^2} = \frac{1}{(2\pi)^3} \int d^3k \, \hbar\omega \tag{14}$$

Since $d^3k/(2\pi)^3$ represents the density of states in momentum space (the number of modes per unit volume in k-space), we can interpret this as:

$$\langle u \rangle = \int \frac{d^3k}{(2\pi)^3} \, \hbar\omega = \int [\text{modes per } k^3] \times [\text{energy per mode}] \tag{15}$$

This confirms that each electromagnetic normal mode (specified by wavevector $\mathbf{k}$ and polarization λ) carries energy $(1/2)\hbar\omega$ in the zero-point radiation field:

$$\boxed{\rho(\omega) = \frac{1}{2}\hbar\omega \quad [\text{energy per mode}]} \tag{16}$$

Comment: I haven't actually verified any of these equations in the last chunk of time ... so we really need to do them by pen soon

4 Maxwell Equations for Gravity

In this section we introduce the second of our two assumptions. The idea that gravity obeys equations analogous to those of electromagnetism was first proposed by Oliver Heaviside in 1893 [**?**], predating general relativity. This is a stronger assumption than it may appear: general relativity is one of the most precisely tested theories in physics. We adopt this framework nonetheless, because it reproduces the key predictions of general relativity to leading order, as shown in **Section 7**. Specifically, we assume that gravity obeys the Maxwell equations with mass and mass currents replacing charge and charge currents as sources. In natural units ($c = 1$):

$$
\begin{aligned}
\boldsymbol{\nabla} \times \mathbf{B} - \partial_t \mathbf{E} = -4\pi G \mathbf{J}, \qquad & \boldsymbol{\nabla} \times \mathbf{E} + \partial_t \mathbf{B} = 0 \\
\boldsymbol{\nabla} \cdot \mathbf{E} = -4\pi G \rho, \qquad & \boldsymbol{\nabla} \cdot \mathbf{B} = 0
\end{aligned} \tag{17}
$$

These equations describe a vector field sourced by mass density ρ and mass current $\mathbf{J}$, with G the gravitational coupling constant.

The continuity equation, although derivable from the above, is worth stating explicitly:

$$
\boldsymbol{\nabla} \cdot \mathbf{J} + \partial_t \rho = 0 \tag{18}
$$

The presence of $\mathbf{B}$, a gravitomagnetic field, may be unexpected. It emerges naturally: just as a moving charge generates a magnetic field, a moving mass generates a gravitomagnetic field. The effect is negligible unless the mass is both extremely large and rapidly moving. **Section 7** discusses a physical manifestation and its potential testability.

The remainder of this section develops mathematical tools that will be used throughout the paper.

4.1 Irrotational and Solenoidal Decomposition

Since any vector field can be split into irrotational (curl-free) and solenoidal (divergence-free) components,

$$
\mathbf{V} = \mathbf{V}_I + \mathbf{V}_S, \quad \boldsymbol{\nabla} \cdot \mathbf{V}_S = 0, \quad \boldsymbol{\nabla} \times \mathbf{V}_I = 0 \tag{19}
$$

We may apply this to our gravitational fields yielding two decoupled subsystems:

Static Fields (Irrotational):

$$
\boldsymbol{\nabla} \cdot \mathbf{E}_I = -4\pi G \rho, \quad \partial_t \mathbf{E}_I = 4\pi G \mathbf{J}_I, \quad \boldsymbol{\nabla} \cdot \mathbf{J}_I + \partial_t \rho = 0 \tag{20}
$$

Radiation Fields (Solenoidal):

$$
\boldsymbol{\nabla} \times \mathbf{B}_S - \partial_t \mathbf{E}_S = 4\pi G \mathbf{J}_S, \quad \boldsymbol{\nabla} \times \mathbf{E}_S + \partial_t \mathbf{B}_S = 0 \tag{21}
$$

This decomposition reflects a meaningful physical distinction between the **static gravitational attraction** and the **propagating radiation** field. Importantly, the energy stored in the irrotational field carries a factor of $1/G$ reflecting the gravitational coupling, whereas the solenoidal radiation field carries energy in the standard electromagnetic sense. This asymmetry will play a role in the energy balance of Section 6.

In the rest frame of the mass, the irrotational and solenoidal subsystems decouple. The irrotational equations govern the static gravitational field sourced by the mass, while the solenoidal equations describe the propagating radiation field. This separation allows us to analyze the radiation field independently of the static gravitational component.

We can make this explicit by defining a new current:

$$\mathbf{J}'_{\mathbf{S}} = -G\mathbf{J}_{\mathbf{S}} \tag{22}$$

This transforms the inhomogeneous solenoidal equation into:

$$\boldsymbol{\nabla} \times \mathbf{B}_S - \partial_t \mathbf{E}_S = 4\pi \mathbf{J}'_{\mathbf{S}} \tag{23}$$

while leaving all other equations unchanged.

Since this form matches the standard presentation in the literature on electromagnetic radiation, we will drop the prime and adopt the following as the solenoidal (radiation) Maxwell equations:

$$\begin{aligned} \boldsymbol{\nabla} \times \mathbf{B}_S - \partial_t \mathbf{E}_S &= 4\pi \mathbf{J}_S \\ \boldsymbol{\nabla} \times \mathbf{E}_S + \partial_t \mathbf{B}_S &= 0 \end{aligned} \tag{24}$$

4.2 Spherical Vector Harmonics

Due to the symmetric radial symmetry of most of the problems associated with gravity, we will benefit from using Vector Spherical Harmonics.

Following Berrara (http://iopscience.iop.org/article/10.1088/0143-0807/6/4/014/meta) (we should send this to the references when we get a chance) we assume that any vector field, $\mathbf{V}$, and any scalar, g, can be expanded as follows:

$$\begin{aligned} \mathbf{V}(\mathbf{x}) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \big[& a_{lm}(r)\, \mathbf{Y}_{lm}(\theta, \phi) + b_{lm}(r)\, \hat{\mathbf{r}} \times \mathbf{X}_{lm}(\theta, \phi) \\ & + c_{lm}(r)\, \mathbf{X}_{lm}(\theta, \phi) \big] \end{aligned} \tag{25}$$

$$g(\mathbf{x}) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} g_{lm}(r) Y_{lm}(\theta, \phi) \tag{26}$$

where Y_{lm} are the usual spherical harmonics (see Jackson 3.53), $\mathbf{Y}_{lm}$ is simply $\hat{\mathbf{r}} Y_{lm}$ and $\mathbf{X}_{lm}$ is defined as,

$$\mathbf{X}_{lm}(\theta, \phi) = \frac{-i}{\sqrt{l(l+1)}} \mathbf{r} \times (\boldsymbol{\nabla} Y_{lm}(\theta, \phi)) \tag{27}$$

4.3 Irrotational Solutions

We now calculate Newtonian gravity as a solution to the irrotational parts of the Maxwell equations.

In the case of a static point mass m located at the origin, we set:

$$\rho(\mathbf{x}) = m\delta^3(\mathbf{x}), \quad \mathbf{J} = 0 \tag{28}$$

Using the irrotational part of the Maxwell equations:

$$\boldsymbol{\nabla} \cdot \mathbf{E}_I = -4\pi G\rho \qquad\qquad \boldsymbol{\nabla} \times \mathbf{E}_I = 0 \tag{29}$$

and using the spherical vector harmonics we developed earlier, we find:

$$\mathbf{E}_I(\mathbf{x}) = \sum_{l,m} \left[-\frac{4\pi i G}{2l+1} \frac{q_{lm}}{r^{l+2}} \sqrt{l(l+1)}\, (\hat{\mathbf{r}} \times \mathbf{X}_{lm}) - \frac{4\pi G(l+1)}{2l+1} \frac{q_{lm}}{r^{l+2}} \mathbf{Y}_{lm} \right] \tag{30}$$

Where,

$$q_{lm} \equiv \int Y_{lm}^*(\theta,\phi) r^l \rho(\mathbf{x})\ d^3x = \int \rho_{lm}(r) r^{l+2}\ dr \tag{31}$$

Clearly then, if

$$\rho(\mathbf{x}) = m\delta^3(\mathbf{x}) \tag{32}$$

We get

$$\mathbf{E}_I(\mathbf{x}) = -\hat{\mathbf{r}} \frac{mG}{r^2} \tag{33}$$

as expected. This confirms that Newtonian gravity is recovered as a special case of the irrotational Maxwell equations , a necessary consistency check for the framework. Note that while the point mass idealization suffices here, in **Section 5** we refine this to a shell distribution (equation 53) to address the gravitational self-energy problem.

4.4 Solenoidal Solutions

In the last subsection, we derived Newtonian gravity from the irrotational equations, in this subsection we will derive the equations of radiation from teh solenoidal solutions. Several results from the solutions of the solenoidal Maxwell equations will be needed later; we now solve the corresponding equations.

Outside the mass, the solenoidal field equations in vacuum are:

$$\boldsymbol{\nabla} \times \mathbf{E}_S + \partial_t \mathbf{B}_S = 0, \quad \boldsymbol{\nabla} \times \mathbf{B}_S - \partial_t \mathbf{E}_S = 0 \tag{34}$$

We write the complete multipole expansion including both electric and magnetic type modes:

$$\mathbf{B} = \int d\omega \sum_{l,m} \left[a_E(l,m) f_l(kr) \mathbf{X}_{lm} - \frac{i}{k} a_M(l,m) \boldsymbol{\nabla} \times (g_l(kr)\mathbf{X}_{lm}) \right] e^{-i\omega t} W_{l,m}(\omega) + \text{c.c.}$$
$$\mathbf{E} = \int d\omega \sum_{l,m} \left[\frac{i}{k} a_E(l,m) \boldsymbol{\nabla} \times (f_l(kr)\mathbf{X}_{lm}) + a_M(l,m) g_l(kr) \mathbf{X}_{lm} \right] e^{-i\omega t} W_{l,m}(\omega) + \text{c.c.}$$
$$\tag{35}$$

where the functions f_l and g_l are linear superpositions of radial Hankel functions $h_l^{(1)}(kr)$ and $h_l^{(2)}(kr)$, and where $W_{l,m}(\omega)$ are complex Gaussian stochastic variables satisfying the same statistical conditions as equation (6), now indexed by (l, m, ω) rather than $(\lambda, \mathbf{k})$:

$$\langle W_{l,m}(\omega)\rangle = 0, \quad \langle W_{l,m}(\omega) W^*_{l',m'}(\omega')\rangle = \delta_{ll'}\delta_{mm'}\delta(\omega - \omega') \tag{36}$$

For a massive particle that absorbs radiation (only incoming waves), we have:

$$f_l(kr) = h_l^{(1)}(kr), \quad g_l(kr) = h_l^{(1)}(kr) \tag{37}$$

4.4.1 Computing the Field Momentum

The total field momentum is:

$$\mathbf{P}_{\text{field}} = \frac{1}{4\pi}\int_V \mathbf{E} \times \mathbf{B}\, d^3x \tag{38}$$

Comment: might be an 8 in that denominator .. but no, I don't think so

Following the math we laid out in **Section 3**, we compute the stochastic average $\langle \mathbf{E} \times \mathbf{B}\rangle$. Writing $\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_+^*$ and $\mathbf{B} = \mathbf{B}_+ + \mathbf{B}_+^*$, the terms that survive the stochastic average are those involving $\langle WW^*\rangle$:

$$\langle \mathbf{E} \times \mathbf{B}\rangle = \langle \mathbf{E}_+ \times \mathbf{B}_+^*\rangle + \langle \mathbf{E}_+^* \times \mathbf{B}_+\rangle \tag{39}$$

Working through the cross products using the orthogonality relations of vector spherical harmonics, and noting that $h_l^{(1)*} = h_l^{(2)}$, the angular integrals over $|\mathbf{X}_{lm}|^2$ yield normalization constants that can be absorbed.

$$\langle \mathbf{P}_{\text{field}}\rangle = -\hat{\mathbf{r}}\int dr \int d\omega \sum_{l,m}\left[\frac{-i}{k}|a_E|^2 h_l^{(2)}\frac{1}{r}\frac{d}{dr}\left(rh_l^{(1)}\right) + \frac{i}{k}|a_M|^2 h_l^{(1)}\frac{1}{r}\frac{d}{dr}\left(rh_l^{(2)}\right)\right] \tag{40}$$

The above can be simplified by use of the Wronskian, leaving:

$$\boxed{\langle \mathbf{P}_{\text{field}}\rangle = -\hat{\mathbf{r}}\int dr \int d\omega \sum_{l,m}\frac{|a_E|^2|\mathbf{X}_{lm}|^2}{4\pi k^2 r^2}} \tag{41}$$

Applying the hypothesis , we identify the spectral momentum density in the integrand:

$$\mathbf{P}(r) = -\hat{\mathbf{r}}\frac{Gm\hbar}{r^2} \tag{1}$$

This allows us to write:

$$\langle \mathbf{P}_{\text{field}}\rangle = -\hat{\mathbf{r}}\int dr \int d\omega \frac{Gm\hbar}{r^2} \tag{42}$$

and therefore:

$$\sum_{l,m} \frac{|a_E|^2 |\mathbf{X}_{lm}|^2}{4\pi k^2} = Gm\hbar \tag{43}$$

4.4.2 Field Energy in the Radiation Zone

The total field energy density is given by:

$$u = \frac{1}{8\pi}(\mathbf{E}^2 + \mathbf{B}^2) \tag{44}$$

Computing the stochastically averaged energy density for the full field is complicated due to cross terms between electric and magnetic multipoles. However, in the radiation zone where $k^2r^2 >> 1$, the fields simplify considerably.

In the radiation zone, using $f_l(kr) \approx h_l^{(1)}(kr) \approx \frac{(-i)^{l+1}}{kr} e^{ikr}$ for incoming waves, the stochastic fields become:

$$\begin{aligned}
\mathbf{B} &= \frac{e^{ikr}}{kr} \int d\omega \sum_{l,m} (-i)^{l+1} \left[a_E(l,m)\mathbf{X}_{lm} + a_M(l,m)\hat{\mathbf{r}} \times \mathbf{X}_{lm} \right] e^{-i\omega t} W_{l,m}(\omega) + \text{c.c.} \\
\mathbf{E} &= \frac{e^{ikr}}{kr} \int d\omega \sum_{l,m} (-i)^{l+1} \left[a_M(l,m)\mathbf{X}_{lm} - a_E(l,m)\hat{\mathbf{r}} \times \mathbf{X}_{lm} \right] e^{-i\omega t} W_{l,m}(\omega) + \text{c.c.}
\end{aligned} \tag{45}$$

Writing $\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_+^*$ and $\mathbf{B} = \mathbf{B}_+ + \mathbf{B}_+^*$, we compute:

$$\langle \mathbf{E}^2 \rangle = 2\langle \mathbf{E}_+ \cdot \mathbf{E}_+^* \rangle, \quad \langle \mathbf{B}^2 \rangle = 2\langle \mathbf{B}_+ \cdot \mathbf{B}_+^* \rangle \tag{46}$$

For $\mathbf{E}_+ \cdot \mathbf{E}_+^*$:

$$\langle \mathbf{E}_+ \cdot \mathbf{E}_+^* \rangle = \frac{1}{k^2r^2} \int d\omega \sum_{l,m} \left[|a_M|^2 |\mathbf{X}_{lm}|^2 + |a_E|^2 |\hat{\mathbf{r}} \times \mathbf{X}_{lm}|^2 \right] \tag{47}$$

where the cross terms vanish due to the orthogonality $\mathbf{X}_{lm} \cdot (\hat{\mathbf{r}} \times \mathbf{X}_{lm}) = 0$. Similarly:

$$\langle \mathbf{B}_+ \cdot \mathbf{B}_+^* \rangle = \frac{1}{k^2r^2} \int d\omega \sum_{l,m} \left[|a_E|^2 |\mathbf{X}_{lm}|^2 + |a_M|^2 |\hat{\mathbf{r}} \times \mathbf{X}_{lm}|^2 \right] \tag{48}$$

Since $|\hat{\mathbf{r}} \times \mathbf{X}_{lm}| = |\mathbf{X}_{lm}|$ (both are unit vectors perpendicular to $\hat{\mathbf{r}}$), and using the neutrality condition $|a_E|^2 = |a_M|^2$ (see Section 2.1), we obtain:

$$\langle \mathbf{E}^2 + \mathbf{B}^2 \rangle = \frac{4}{k^2r^2} \int d\omega \sum_{l,m} |a_E|^2 |\mathbf{X}_{lm}|^2 \tag{49}$$

The total field energy is therefore:

$$\boxed{\langle U\rangle = \int_V d^3x \int d\omega \sum_{l,m} \frac{|a_E|^2|\mathbf{X}_{lm}|^2}{4\pi k^2 r^2}} \tag{50}$$

The expansion of this calculation outside the radiation zone involves more intricate mathematics. We defer this more general treatment to future work.

5 A Toy Model

Sections 3 and 4 laid out the two assumptions of the theory, both grounded in existing literature. From here, we enter new territory. In this section we introduce a simple model of a massive particle, making several idealizing assumptions in order to gain traction on the problem. Less idealized models are left for future work.

We proceed in two steps. First, working in the space domain, we follow Rohrlich to establish a lower bound on the particle radius. Then, moving to the frequency domain, we introduce a single-frequency absorption model and show that the energy absorbed from the ZPR field is less than what is available, confirming the energetic consistency of the hypothesis.

5.1 Lower Bound: Minimum Radius via Gravitational Self-Energy

We follow Rohrlich[?] closely here, as his framing captures the essential point precisely. The passage has been lightly adapted from the original to reflect that we are discussing massive particles rather than charged ones.

We wish to construct a model of a massive particle. "The most obvious model of a massive particle is a sphere carrying a spherically symmetrical mass distribution. While such a model is meant (and was indeed proposed) as a picture of a massive elementary particle (a neutron, for example), it is obvious that is is basically a macroscopic massive body, only much smaller. There is nothing 'elementary' about it. To see this clearly, let us consided a macroscopic massive sphere in more detail.

"Consider a sphere of radius R and spherically symmetric mass density $\rho(\mathbf{x})$. Its total mass is

$$m = \int_V \rho(\mathbf{x}) d^3x \tag{51}$$

"The integral extending over the volume of the sphere. Since $m \neq 0$ by assumption, there must exist attractive gravitational forces between 'parts' of the total mass. Symmetry would therefore require that this massive sphere contract and continue to do so to arbitrarily small radius R." Thus with no other forces present, the particle will necessarily collapse.

For the purposes of this paper, the mass distribution is taken to be a uniform spherical shell at radius R:

$$\rho(\mathbf{x}) = \frac{m}{4\pi r^2}\delta(r - R) \tag{52}$$

For brevity, we write this as:

$$\rho(\mathbf{x}) = m\,\delta_{\text{shell}}(\mathbf{x}) \tag{53}$$

In the next section, we show that the particle need not collapse. For now, we take it as given that it does not, and ask what gravitational self-energy implies for the radius.

Since no forces other than gravity are present, the total energy satisfies:

$$U = m = -\frac{1}{2}\int \frac{Gm\rho(\mathbf{x})}{r} d^3x \tag{54}$$

The more fundamental relativistic statement is $U^2 = m^2$ (equivalently, $p_\mu p^\mu = m^2$), which we adopt throughout.

Choosing the simplest mass distribution, a uniform spherical shell at radius R, the self-energy integral yields:

$$R = \frac{Gm}{2} \tag{55}$$

Other choices of mass distribution (a uniform sphere, for example) give R of the same order. We therefore take $R = Gm$ as the characteristic minimum radius:

$$\boxed{R = Gm} \tag{56}$$

This is analogous in spirit to the classical electron radius, where self-energy sets a length scale.

5.1.1 Compactness of Physical Objects

It is illuminating to compare the mathematical radius with the radii of real physical bodies. Slipping back to SI units, we tabulate the compactness parameter $\Phi = \frac{Gm}{Rc^2}$ for several objects:

$$\text{Mathematical limit:} \quad \Phi = 1$$
$$\text{Black hole:} \quad \Phi = 1$$
$$\text{Sun:} \quad \Phi = 2 \times 10^{-6}$$
$$\text{Earth:} \quad \Phi = 7 \times 10^{-10}$$
$$\text{Proton:} \quad \Phi = 1 \times 10^{-38}$$

Notably, the mathematical limit $\Phi = 1$ coincides with the black hole threshold, suggesting that $R = Gm$ is not merely a formal bound but a physically meaningful one.

5.2 Frequency Domain: The Single-Mode Approximation

Boyer showed rigorously that a dipole oscillator in the ZPR field absorbs from a remarkably narrow frequency band [**?**]; our single-frequency model can be understood as the idealized limit of this result. Any real model of a macroscopic massive object that absorbs radiation will absorb over a non-singular domain. However, to gain traction, we adopt a toy model that absorbs at only one frequency. We choose:

$$k_\circ = \omega_\circ = \frac{m}{\hbar} \tag{57}$$

In spherical coordinates, this corresponds to the frequency distribution:

$$\frac{\delta\left(\omega - \frac{m}{\hbar}\right)}{4\pi\omega^2} \tag{58}$$

The energy absorbed from the field is then, using (16):

$$U = \int d\omega\, \hbar\omega\, \delta\left(\omega - \frac{m}{\hbar}\right) = m \tag{59}$$

To show that the particle draws less energy than is available from the ZPR, we introduce a multiplicative factor α. The self-energy condition gives $U = Gm^2/R$, and the absorbed energy is:

$$U = \alpha \int d\omega\, \hbar\omega\, \delta\left(\omega - \frac{m}{\hbar}\right) \tag{60}$$

Combining these:

$$\alpha = \frac{Gm}{R} \tag{61}$$

Since we established in the previous section that $R > Gm$, we have $\alpha < 1$: the particle absorbs less energy from the ZPR than is available. The toy model is energetically consistent.

5.3 On-Shell Conditions

Before moving on, we collect the defining quantities of the model and show they are mutually consistent. We have the radius (56), the absorbed frequency, and the hypothesis (1) evaluated at $r = R$:

$$R = Gm, \qquad k_\circ = \frac{m}{\hbar}, \qquad p = \frac{Gm\hbar}{R^2} \tag{62}$$

Combining these:

$$\frac{Gm\hbar}{R^2} = \hbar k_\circ = m \tag{63}$$

Multiplying through by $R/\hbar$:

$$\frac{Gm}{R} = k_\circ R = 1 \tag{64}$$

and therefore:

$$\boxed{k_\circ^2 R^2 = 1} \tag{65}$$

This result places the model exactly at the boundary between the near and far field regimes — neither $k_\circ R \ll 1$ nor $k_\circ R \gg 1$. We refer to this as being *on-shell*. Throughout what follows, $\delta^{(3)}_{\text{shell}}(\mathbf{x})$ denotes that we never probe scales smaller than R.

6 Field Energy and Momentum Balance: A Mechanism for Stability

Now that we have established our model of a massive particle in the previous section, namely:

$$R = Gm, \qquad m = \hbar k_\circ, \qquad k_\circ R = 1 \tag{66}$$

we are ready to examine whether the proposed inflow of zero-point radiation can stabilize a massive particle against gravitational collapse. The instability in question was introduced in Section 5: a self-gravitating mass has divergent self-energy and will collapse unless some mechanism intervenes. We propose that the inflow of ZPR radiation provides exactly this mechanism.

Following Rohrlich, if the energy absorbed from the ZPR can offset the divergence in the gravitational self-energy, the particle is stable. We first analyze the energy density and total energy of the system, then examine whether this balance leads to equilibrium. Finally, we turn to momentum conservation, where we derive an effective "Ohm's law" relating solenoidal fields and currents.

6.1 Stability from Field Energy

We now evaluate whether the inward radiation flow proposed in **Section 2.1** can stabilize a massive particle against gravitational collapse. Specifically, we ask whether the energy absorbed from the ZPR can counterbalance the gravitational self-energy that would otherwise drive collapse.

As established in **Section 4.1**, the irrotational and solenoidal fields carry energy differently. The irrotational contribution is:

$$U_I = \frac{1}{4\pi G} \int |\mathbf{E}_I|^2 \, d^3x \sim \int \frac{Gm^2}{r^4} \, d^3x \tag{67}$$

Comment: verify the E_I integral in next pass

This integral diverges, confirming that a massive particle would collapse under its own gravitational self-energy without a stabilizing mechanism. The radiation field provides precisely this stabilization. Including both contributions, the total energy is:

$$U_{\text{tot}} = \int_V \left(-\frac{1}{4\pi G} |\mathbf{E}_I|^2 + |\mathbf{E}_S|^2 + |\mathbf{B}_S|^2 \right) d^3x \tag{68}$$

From equation (50), and applying the hypothesis, the radiation contribution is:

$$U_{\text{rad}} = \frac{1}{8\pi^3} \int \int \frac{Gm\hbar}{r^2} \, d^3x \, d^3k$$

Integrating over k and suppressing numerical prefactors, which are absorbed into the definition of the model (as in **Section 5**), the total energy becomes:

$$U_{\text{tot}} = \int \left[\frac{Gm\hbar}{r^2} k^3 - \frac{Gm^2}{r^4} \right] d^3x \tag{69}$$

Evaluating on-shell, with $k = m/\hbar$ and $r = R = Gm$, both terms are equal and the integrand vanishes:

$$U_{\text{tot}} = 0 \tag{70}$$

The total energy vanishes on-shell, demonstrating that the radiation inflow exactly counterbalances the gravitational self-energy. The massive particle is stable.

6.2 Momentum Conservation and an Effective Ohm's Law

We now turn to momentum conservation. The total momentum exchange in the system must vanish:

$$\int_V \left(\rho \mathbf{E}_I + \mathbf{J}_S \times \mathbf{B}_S \right) d^3x = 0 \tag{71}$$

We assume the solenoidal current obeys a linear response:

$$\mathbf{J}_S = \sigma \mathbf{E}_S \tag{72}$$

Using our on-shell model (see (53)), with $\rho = m\delta^3(\mathbf{x})$, and averaging over the stochastic field, the momentum balance becomes:

$$-\hat{\mathbf{r}} \int \left(m\delta^3(\mathbf{x}) \cdot \frac{Gm}{r^2} - \sigma \cdot \frac{Gm\hbar}{r^2} \right) d^3x = 0 \tag{73}$$

Solving for σ:

$$\sigma = \frac{m}{\hbar} \delta^3(\mathbf{x}) \tag{74}$$

This is a delta-function conductivity concentrated at the origin, or more precisely, at the shell $r = R$. It defines an effective Ohm's law in which the mass acts as a perfect point-like absorber for incoming radiation outside the radius R. The relevant Maxwell equation becomes:

$$\boldsymbol{\nabla} \times \mathbf{B}_S - \partial_t \mathbf{E}_S = -\frac{m}{\hbar}\mathbf{E}_S \delta^3(\mathbf{x}) \tag{75}$$

Together, the energy and momentum analyses of this section demonstrate that the hypothesis is internally consistent: a massive particle that sinks zero-point radiation is stabilized against gravitational collapse by drawing energy from the vacuum field itself.

7 Classical Tests of Gravity

This section demonstrates that the present theory reproduces the three classic experimental tests of general relativity, first proposed by Einstein [**?**]: the precession of Mercury's perihelion, the deflection of light near a massive object, and gravitational redshift.

7.1 Precession of the Perihelion of Mercury

In electrodynamics, a static Coulomb field transforms under Lorentz boosts into a configuration with both electric and magnetic components. Specifically, the fields of a charge moving at velocity $\mathbf{v}$ are:

$$\mathbf{E}' = \gamma \mathbf{E} - \frac{\gamma^2}{\gamma+1}\mathbf{v}(\mathbf{v}\cdot\mathbf{E}), \quad \mathbf{B}' = -\gamma \mathbf{v} \times \mathbf{E} \tag{76}$$

where $\gamma = 1/\sqrt{1-v^2}$.

By analogy, we assume the same Lorentz transformation laws apply to gravity. We define the gravitational field:

$$\mathbf{E}_g = \mathbf{F}/m = -\frac{GM}{r^3}\mathbf{r} \tag{77}$$

Then, under a Lorentz boost:

$$\mathbf{E}'_g = \gamma \mathbf{E}_g - \frac{\gamma^2}{\gamma+1}\mathbf{v}(\mathbf{v}\cdot\mathbf{E}_g), \quad \mathbf{B}'_g = -\gamma \mathbf{v} \times \mathbf{E}_g \tag{78}$$

In the low-velocity limit $v \ll 1$, we find:

$$\mathbf{B}_g = -\mathbf{v} \times \mathbf{r} \cdot \frac{GM}{r^3} \tag{79}$$

This introduces a "gravitomagnetic" interaction analogous to magnetism in electrodynamics.

We now consider Kepler's problem. The Newtonian effective potential is:

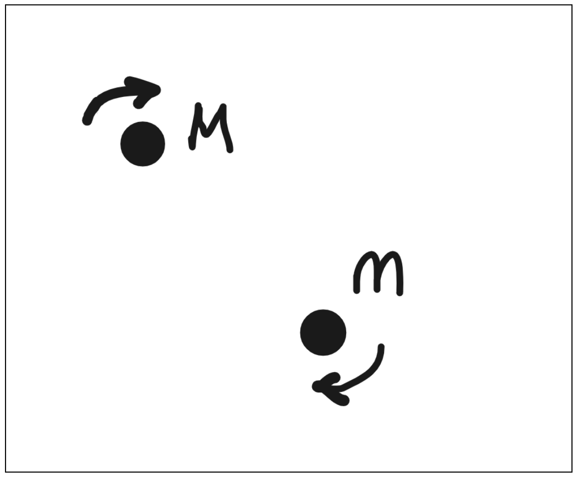

Figure 1: Two masses in mutual orbit

$$U_{\text{Newt}} = \frac{1}{2}m\left(\frac{dr}{dt}\right)^2 - \frac{GMm}{r} + \frac{l^2}{2mr^2} \tag{80}$$

where $l = m|\mathbf{r} \times \mathbf{v}|$ is the orbital angular momentum (see Figure 1).

A moving charge induces a magnetic moment $\mathbf{m} = \frac{1}{2}e(\mathbf{r} \times \mathbf{v})$. Replacing $e \to m$, we define a gravitational magnetic moment:

$$\mathbf{m}_g = \frac{1}{2}m(\mathbf{r} \times \mathbf{v}) \tag{81}$$

The potential energy of this moment in a magnetic field is:

$$U = -\mathbf{m}_g \cdot \mathbf{B}_g = -\frac{GMm}{2}|\mathbf{r} \times \mathbf{v}|^2/r^3 \tag{82}$$

This adds a correction to the potential energy. Including the reciprocal effect (from the motion of the central mass), we double this term and find the corrected potential:

$$U_{\text{Einstein}} = \frac{1}{2}m\left(\frac{dr}{dt}\right)^2 - \frac{GMm}{r} + \frac{l^2}{2mr^2} - \frac{GMl^2}{mr^3} \tag{83}$$

This is the same correction obtained in general relativity to first order. As Carroll[**?**] notes , general relativity produces this result exactly; in our theory, it emerges as a leading-order approximation. The higher-order terms in our expansion are not present in general relativity, and in principle offer an observational test that could distinguish between the two frameworks. We leave the computation of these higher-order terms for future work.

7.2 Deflection of Light

We now consider the deflection of light near a massive object. Our central hypothesis is that mass creates a momentum-transfer field:

$$\mathbf{P}(r) = -\hat{\mathbf{r}}\frac{Gm\hbar}{r^2} \tag{1}$$

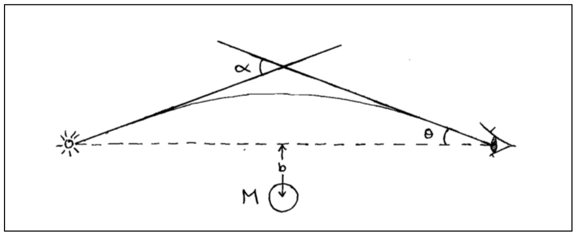

Figure 2: Deflection of light

A photon traveling through this field receives a cumulative momentum shift:

$$\Delta\mathbf{p} = \int_{\text{path}} \mathbf{P}(r)\, dl \tag{2}$$

Assuming a straight-line path passing at minimum distance b (see figure 2), we integrate:

$$\Delta\mathbf{p} = \int_{-\infty}^{\infty} -\frac{Gm\hbar}{r^3}\mathbf{r}\, w\, dz \tag{84}$$

The parallel momentum shifts cancel symmetrically. The transverse component yields:

$$\Delta p_y = \hbar w \cdot \frac{2Gm}{b} \tag{85}$$

The angle of deflection is then:

$$\theta \approx \frac{\Delta p_y}{p_z} = \frac{2Gm}{b} \quad \Rightarrow \quad \alpha = \frac{4Gm}{b} \tag{86}$$

This matches, to first-order, the deflection angle predicted by general relativity.

7.3 Gravitational Redshift

Finally, we consider the redshift of light climbing out of a gravitational well. As before, a photon gains or loses momentum depending on its path through the momentum field.

Using:

$$\Delta p = \int_{\infty}^{R} \mathbf{P} \cdot \hat{\mathbf{r}} \cdot w \, dr = \int_{\infty}^{R} -\frac{Gm\hbar}{r^2} w \, dr = \hbar w \cdot \frac{Gm}{R} \tag{87}$$

The resulting frequency shift is:

$$\hbar w' = \hbar w + \Delta p = \hbar w \left(1 + \frac{Gm}{R}\right) \tag{88}$$

which, to first order, matches the prediction of general relativity for gravitational redshift.

8 Duality and the Dirac Equation

Throughout this paper, we have stayed grounded in classical physics. We built a theory of gravity using field equations, conservation laws, and an underlying sea of vacuum radiation. No quantum assumptions, no curved spacetime. In this section, something surprising appears. When the field equations are rewritten in a new form, a familiar structure emerges: the Dirac equation, the cornerstone of quantum mechanics.

8.1 Duality Transformations

As discussed in Jackson [?] (Section 6.11), the Maxwell equations can be extended to include both electric and magnetic charges:

$$\begin{aligned} \boldsymbol{\nabla} \times \mathbf{B} - \partial_t \mathbf{E} &= \mathbf{J}_e, \qquad & \boldsymbol{\nabla} \times \mathbf{E} + \partial_t \mathbf{B} &= -\mathbf{J}_m \\ \boldsymbol{\nabla} \cdot \mathbf{E} &= \rho_e, & \boldsymbol{\nabla} \cdot \mathbf{B} &= \rho_m \end{aligned} \tag{89}$$

The magnetic densities are assumed to obey the same continuity equation as their electric counterparts.

Now consider the following duality transformation:

$$\begin{aligned} \mathbf{E} &= \mathbf{E}' \cos\xi + \mathbf{B}' \sin\xi, \qquad & \mathbf{B} &= \mathbf{B}' \cos\xi - \mathbf{E}' \sin\xi \\ \rho_e &= \rho_e' \cos\xi + \rho_m' \sin\xi, & \rho_m &= \rho_m' \cos\xi - \rho_e' \sin\xi \\ \mathbf{J}_e &= \mathbf{J}_e' \cos\xi + \mathbf{J}_m' \sin\xi, & \mathbf{J}_m &= \mathbf{J}_m' \cos\xi - \mathbf{J}_e' \sin\xi \end{aligned} \tag{90}$$

This transformation leaves the generalized Maxwell equations invariant. As Jackson [?] notes, the distinction between electric and magnetic charge is, in this framework, a matter of convention. If all particles share the same magnetic-to-electric charge ratio, a duality rotation can eliminate magnetic charges entirely. This perspective opens the door for deeper reinterpretations of classical field theory.

8.2 Spinor Representation of Field Operators

We now examine a way to recast Maxwell's equations using spinor-like objects. Define the vector curl as a matrix operator:

$$\boldsymbol{\nabla} \times \mathbf{V} = g_i \partial_i \mathbf{V} \tag{91}$$

with:

$$g_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad g_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad g_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{92}$$

These are the standard generators of SO(3) rotations, the antisymmetric 3×3 matrices representing infinitesimal rotations in three-dimensional space. The representation $\boldsymbol{\nabla} \times \mathbf{V} = g_i \partial_i \mathbf{V}$ can be verified by direct computation.

We define the gamma matrices as:

$$\gamma^0 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & g_i \\ -g_i & 0 \end{pmatrix} \tag{93}$$

These matrices satisfy the algebra of infinitesimal rotations and generate the curl operator in 3D. They are not the standard Dirac representation, which operates on four-component spinors in Minkowski space, but they serve to make the structural similarity to the Dirac equation manifest. This connection could be further explored with tools from representation theory and Clifford algebras.

8.3 Emergence of the Dirac Equation

Recall from Section 6 that the on-shell momentum conservation equation (75) gives:

$$\boldsymbol{\nabla} \times \mathbf{B}_S - \partial_t \mathbf{E}_S = -\frac{m}{\hbar} \mathbf{E}_S \delta^3(\mathbf{x}) \tag{75}$$

Restricting to the on-shell condition (see Section 5), we drop the delta function:

$$\boldsymbol{\nabla} \times \mathbf{B}_S - \partial_t \mathbf{E}_S = -\frac{m}{\hbar} \mathbf{E}_S \tag{94}$$

Applying the duality transformation, the solenoidal Maxwell equations become:

$$\begin{aligned} -\partial_t \mathbf{E}_S + \boldsymbol{\nabla} \times \mathbf{B}_S + \frac{m}{\hbar} \mathbf{E}_S &= 0 \\ -\partial_t \mathbf{B}_S - \boldsymbol{\nabla} \times \mathbf{E}_S + \frac{m}{\hbar} \mathbf{B}_S &= 0 \end{aligned} \tag{95}$$

Using the spinor curl definition:

$$
\begin{aligned}
-\partial_t \mathbf{E}_S + g_i \partial_i \mathbf{B}_S + \frac{m}{\hbar}\mathbf{E}_S = 0 \\
-\partial_t \mathbf{B}_S - g_i \partial_i \mathbf{E}_S + \frac{m}{\hbar}\mathbf{B}_S = 0
\end{aligned}
\tag{96}
$$

Defining a six-component field:

$$
\psi = \begin{pmatrix} \mathbf{E}_S \\ \mathbf{B}_S \end{pmatrix}
\tag{97}
$$

these two equations combine into:

$$
\left(\gamma^\mu \partial_\mu + \frac{m}{\hbar} \right) \psi = 0
\tag{98}
$$

This has the exact form of the Dirac equation, as given in Sakurai's Advanced Quantum Mechanics[**?**], Eq. 3.31.

This is a remarkable result: starting from classical field equations and a single hypothesis, we arrive at a structure identical to the Dirac equation, without quantization or any reference to spin. The six-component field $\psi = (\mathbf{E}_S, \mathbf{B}_S)^T$ is a classical field object, not a quantum spinor. The quantum mechanical structure emerges from the geometry of the field equations themselves.

9 Conclusion

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Can gravity emerge from classical physics, without invoking curved spacetime or quantum gravity? That was the question we asked. Faced with two powerful but incompatible theories, quantum mechanics and general relativity, we chose a different path: to treat mass as a source of momentum flow, drawing from a pervasive background field. The result is a framework that is simple, classical, and deeply physical.

To test the idea, we built the theory step by step. We formulated equations for how mass interacts with the vacuum field. We introduced cutoffs to make the interaction finite and consistent. We checked for conservation of energy and momentum. Then we asked: does this theory match observation? And the answer, to leading order, is yes. It reproduces the familiar predictions of general relativity: the bending of light, gravitational redshift, and the precession of Mercury. We also identify a testable difference between General Relativity and the theory presented here.

But something unexpected happened. When we re-expressed the field equations in a different form, they gave rise to the Dirac equation, the same equation that governs quantum particles like electrons. This wasn't assumed; it emerged. That a single classical field framework can describe gravity, electromagnetism, and quantum behavior is remarkable. It suggests these aren't separate layers of physics, but different faces of the same underlying structure. One set of equations that does it all.

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This stands in contrast to general relativity, where mass is treated as a cold object sitting in empty space and curving spacetime around it

9.1 Where This Could Go

Here we simply state the hypothesis and explore its consequences. Several directions remain open for future work:

Derivation of the hypothesis. The idea that mass sinks radiation appears explicitly in Feynman's lectures on gravity *Comment: verify exact quote and citation*. What is novel here is the specific form of the momentum density, $\mathbf{P}(r) = -\hat{\mathbf{r}}Gm\hbar/r^2$. It is the belief of this author that this expression can be derived directly from the two assumptions of the paper, rather than postulated. That derivation is left for future work.

Derivation of the hypothesis. The intuition underlying the hypothesis has respectable precedent. Feynman notes that light "falls" in a gravitational field [**?**], which is precisely the picture developed here: mass acts as a sink for radiation. What is novel is the specific form of the momentum density, $\mathbf{P}(r) = -\hat{\mathbf{r}}Gm\hbar/r^2$. It is the belief of this author that this expression can be derived directly from the two assumptions of the paper, rather than postulated. That derivation is left for future work.

Energy expansion outside the radiation zone. The energy calculation in Section 6 was carried out only in the radiation zone. Extending this to the full domain involves richer mathematics and, we suspect, richer physics.

Higher-order perihelion precession. The perihelion precession calculation was performed to leading order, matching general relativity. Higher-order terms in our expansion are not present in general relativity and could provide an observational test distinguishing the two frameworks.

Less idealized models. The toy model of a massive particle adopted here is as simple as possible. Less idealized models would be worth studying in future work.

Experimental test. We identified a testable difference between this theory and general relativity. Designing a concrete experiment to probe this difference is an interesting open challenge.

9.2 Final Thought

Physicists have long sought simplicity. Apparently the universe is built on simple laws. Einstein advised making a theory as simple as possible, but no simpler. When a single framework explains more with less, it earns our attention. Bringing electricity, magnetism, gravity, and the Dirac equation under one set of equations is a level of simplicity that deserves serious attention.

The history of physics is full of bold guesses. Some are wrong, some are premature, and some are foundational. This is just a guess. But it is a specific, testable, and mathematically structured one. If nothing else, it shifts the question. Gravity need not emerge from geometry. It might instead arise from the quiet hiss of radiation no one thought to listen to.

This is offered not as a conclusion, but as a beginning. Collaborators welcome.

Acknowledgments

This paper would not exist without the support of a few key people. First and foremost, I thank my wife, who has stood beside me for over two decades as I pursued this idea. Her patience, steadiness, and quiet encouragement through all the ups and downs have been unwavering.

I am also deeply grateful to my friend Michael Seidel, who, despite not having a background in physics, listened to me explain these ideas again and again over the years. His willingness to engage, ask questions, and simply be present helped me find clarity when I needed it most.

To both of them: thank you.

References